

DESIGN NOTES

LoRa-Based Pet Tracking Collar

Antenna Placement · Power Budget · Production (1,000-Unit) Changes

1. Antenna Placement Rationale

The board carries two RF antennas that must be isolated from each other and from lossy or conductive material nearby.

- **GPS antenna (AE2, passive chip antenna, feeding MAX-M10S U2):** needs a clear view of sky and continuous ground-plane clearance per the vendor's keep-out footprint. It should sit on the top / outward-facing side of the collar, away from the dog's neck and fur, since body tissue and moisture significantly detune and attenuate GNSS signal. Keep it away from the battery (BT1) and any metal snap/buckle hardware, both of which are strong RF absorbers at these dimensions.
- **868 MHz LoRa antenna (E1, ANT-868-CHIP-T, feeding RAK3172 U1):** ceramic chip antennas are sensitive to ground-plane length and shape — follow the datasheet's reference layout exactly, with the antenna at a board edge, radiating away from the PCB, battery, and GPS antenna.
- **Antenna-to-antenna spacing:** place the GPS and LoRa antennas at opposite ends of the PCB to minimize near-field coupling and de-sensing between the two radios operating simultaneously.
- **Body-worn effects:** orient both antennas so the strongest lobe points away from the animal's body, keeping the metal battery cell and any D-ring/buckle outside each antenna's near-field — the most common cause of range loss in wearable trackers.

2. Power Budget Estimate

Assumes a 300 mAh, 3.7 V Li-Po cell.

State	Dominant load	Approx. current
Sleep	MCU standby, LoRa & GPS off, accelerometer wake-on-motion, regulator quiescent draw	~5–8 mA (AMS1117) / ~10–50 μ A (low-IQ)
GPS ON	MAX-M10S GNSS acquisition + fix	~20–25 mA
LoRa TX burst	RAK3172 PA drive, ~14 dBm typical up to +22 dBm boost, 0.1–1 s/packet	~20–40 mA (14 dBm) – ~100–120 mA (22 dBm)

Illustrative duty cycle — one GPS fix + one LoRa uplink every 10 minutes (30 s GPS-on, 1 s TX, remainder sleep):

- With the AMS1117 LDO in the current design (always-on, ~5 mA quiescent): average current $\approx$ 6 mA $\rightarrow$ ~50 hours (~2 days) on 300 mAh.
- With a low-IQ regulator (μ A-class quiescent current): average current $\approx$ 1.3 mA $\rightarrow$ ~225 hours (~9–10 days) on the same cell.

Takeaway: *the always-on linear regulator, not the GPS or radio, is the single biggest lever on standby battery life.*

3. Changes for a 1,000-Unit Production Run

- **Power path:** replace the AMS1117 LDO with a nanopower buck converter or a low-IQ LDO (sub- μ A to low- μ A quiescent) to remove the dominant sleep-current drain identified above.
- **Battery safety:** add a dedicated battery protection IC (over-discharge / over-current / short-circuit) — the TP4056 alone only manages charging, not discharge protection, which is a field-reliability and safety gap at volume.
- **BOM cost at scale:** RAK3172 and MAX-M10S module pricing drops materially at 500–1,000+ unit reel quantities versus the single-unit prices used in the current BOM — negotiate reel/tape pricing directly with the module vendor or an authorized distributor.
- **Programming interface:** consider removing the USB-C/CH340E section from the production PCB (or making it a do-not-populate option) and programming/flashing via a pogo-pin test jig at assembly — saves BOM cost and board area across 1,000 units, unless end-user USB firmware updates are a product requirement.
- **RF certification:** the 868 MHz LoRa link and GNSS antenna placement need to go through ETSI (or applicable regional) radio certification before a production run — schedule this early since it can drive PCB/antenna layout changes.
- **Enclosure & sourcing:** move from a prototype/3D-printed enclosure to an injection-molded, IP-rated housing (a neck-worn outdoor device needs water/dust ingress protection), and source the PCB/assembly through a volume-capable contract manufacturer with panelized SMT assembly.